

123456

A

Pi header (5 V, GPIOs) and 3.3 V

File: lcd_dsi_power.kicad_sch

Raspberry Pi DSI input, ICN6211 bridge

File: lcd_dsi_dsi.kicad_sch

B

AT070TN92 panel, bias supplies, VCOM, backlight, touch

File: lcd_dsi_panel.kicad_sch

C

AT070TN92 7" 800x480 TTL panel driver for the Raspberry Pi DSI port: ICN6211 DSI->RGB888 bridge.

J1 = Pi 15-pin DSI (Pi 3/4; Pi 5 / CM via the standard 22->15 pin display cable). Pins 14/15 (3V3) unused.

J2 -> Pi 40-pin header: 5 V (pins 2/4), GND, DSI_EN, TP_INT, TP_RST, BL_PWM GPIOs (see rpi_dsi/).

Touch (GT915) and ICN6211 I2C share the DSI-connector I2C bus (i2c_csi_dsi).

DSI_EN high: panel RESET released after ~1 ms, bias (AVDD/VGL/VGH) after ~50 ms (RC).

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP-contact 0.5 mm 50-pin FPC connector.

D

ICN6211 DSI->RGB888, TPS61040 bias, TPS61165 backlight, GT915 touch on Pi I2C

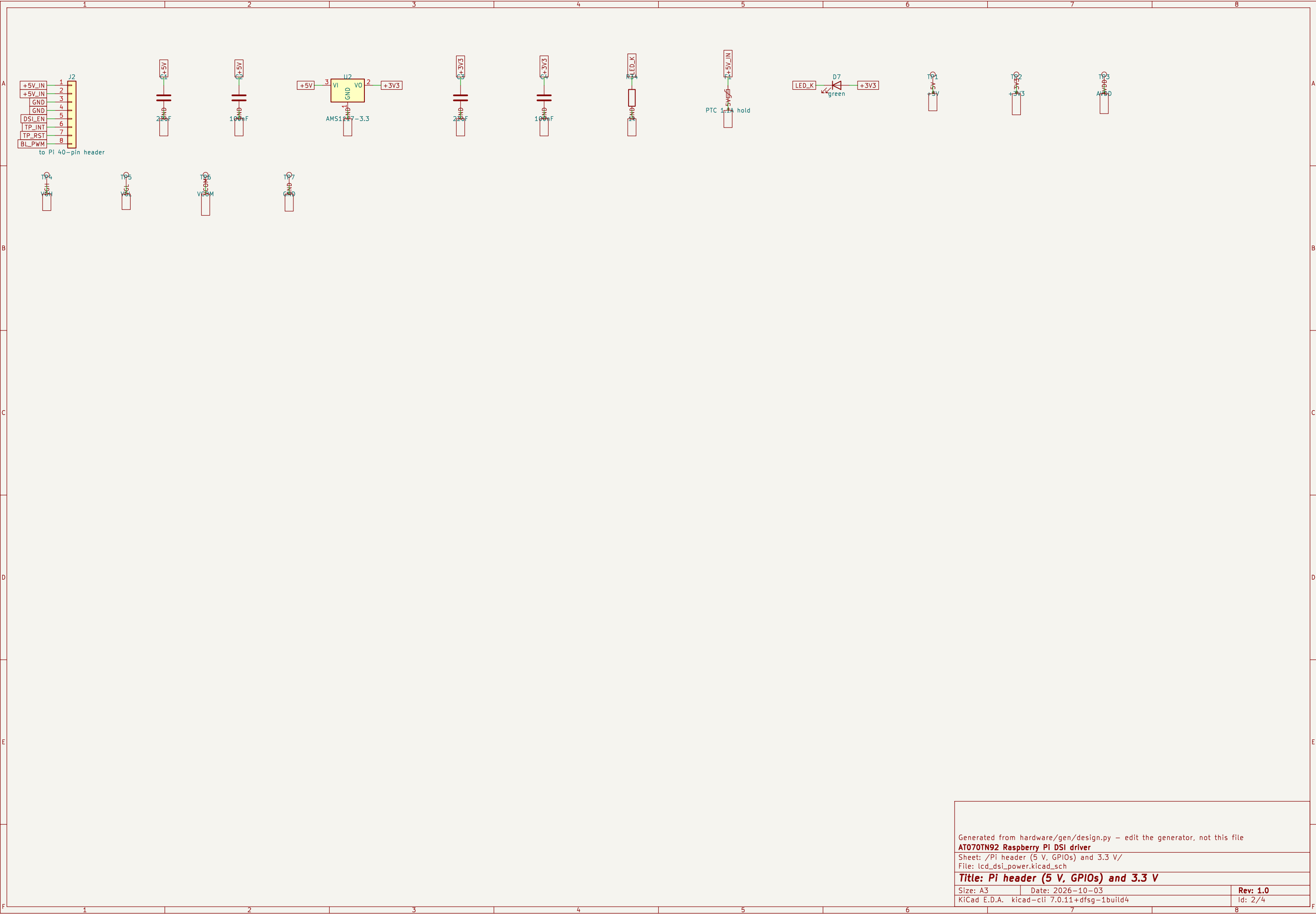
Sheet: /
File: lcd_dsi.kicad_sch

Title: AT070TN92 Raspberry Pi DSI driver board

Size: A4Date: 2026-10-03Rev: 1.0

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/4

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AT070TN92 Raspberry Pi DSI driver

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Size: A3 Date: 2026-10-03

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KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 2/4

